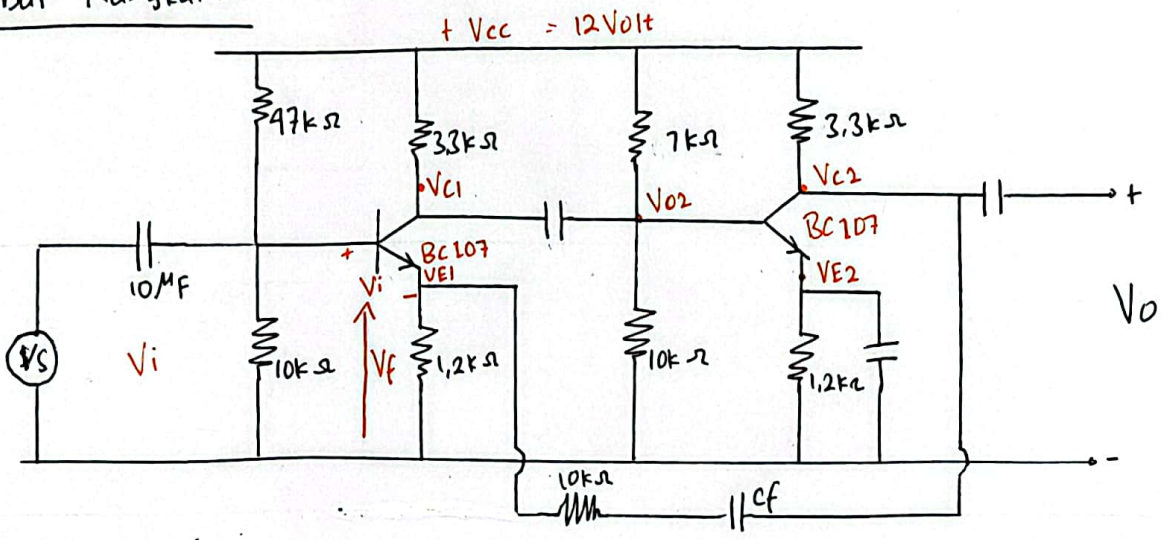
Rangkaian Penguat dengan Tumpan Balik Voltage Seri.

Tanggal Praktikum
09. November 2023

Nama : • Ahsanuamala Asma A (221331002)
• Chistian Adi Pamungkas (221331005)
• Devita Syifa Azzahra (221331009)
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Kelas : 2A - TTE

-> Gambar Rangkaian



-> Hasil Pengukuran

Tegangan / Resistor	V_{B1}	V_{B2}	V_{C1}	V_{C2}	V_{E1}	V_{E2}
10kΩ	2,110V	2,117V	8,76V	7,33V	1,505V	1,515V
100kΩ	2,095V	2,103V	7,39V	7,41V	1,483V	1,491V

-> Hasil Perhitungan

• 10 kΩ

$V_{CE_1} = V_{C_1} - V_{E_1} = 8,76 - 1,505 = 7,255V$

$V_{CE_2} = V_{C_2} - V_{E_2} = 7,33 - 1,515 = 5,815\ V$

$V_{CE\,cutoff} = V_{CC} = 12\ V$

$I_{C_1} = \frac{V_{E_1}}{R_{E_1}} = \frac{1,505}{1,2k} = 1,25\,mA$

$I_{C_2} = \frac{V_{E_2}}{R_{E_2}} = \frac{1,515}{1,2k} = 1,26\,mA$

$I_{C\,sat} = \frac{V_{CE\,cutoff}}{R_C + R_E} = \frac{12}{3,3k + 1,2k} = 2,67\,mA$

• 100 kΩ

$V_{CE_1} = V_{C_1} - V_{E_1} = 7,39 - 1,483 = 5,907V$

$A_{V_f} = \frac{V_O}{V_S} = \frac{9000\,mV_{pp}}{2700\,mV_{pp}} = 3,33$

$V_{CE_2} = V_{C_2} - V_{E_2} = 7,41 - 1,491 = 5,919\ V$

$V_{CE\,cutoff} = V_{CC} = 12\ V$

$I_{C_1} = \frac{V_{E_1}}{R_{E_1}} = \frac{1,483}{1,2k} = 1,235\ mA$

$I_{C_2} = \frac{V_{E_2}}{R_{E_2}} = \frac{1,491}{1,2k} = 1,2425\ mA$

$I_{C\,sat} = \frac{V_{CE\,cutoff}}{R_C + R_E} = \frac{12}{3,3k + 1,2k} = 2,67\ mA$

$$\frac{V_O}{V_i} = \frac{1,667 \cdot \sqrt{2}}{0,744 \cdot \sqrt{2}} = \frac{2,357}{1,05} = 2,245$$

